Symbolic Explainable Artificial Intelligence 2026/27

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Problem

CSPLib `prob039`: The Talent Scheduling Problem (Rehearsal Problem): scenes must be ordered so that the cost of actors waiting on set is minimised. Each actor is paid from their first to their last scene, but only the idle (waiting) days in between count as cost. Our model encodes the schedule as a permutation of scenes, with first/last slot variables per actor and a prefix sum of durations for the cost. It was written from scratch from the problem specification, with symmetry breaking (seq[1] $<$ seq[n]) and a redundant span constraint (Smith, 2003).

Extension

Each scene has a filming location, and moving between locations has a cost given by a (possibly asymmetric) matrix `travel[A, B]`. The extended model minimises actor waiting cost + travel cost. The two goals conflict: on `tradeoff_demo` the actor-only optimum (cost 0) costs 110 in travel, whereas the joint optimum is 85 ($55 + 30$). The reversal symmetry breaking valid for the base model is not valid with asymmetric travel costs, so it was removed from the extended model. With an all-zero travel matrix the extended model gives the same optimum as the base model (Film2: 87).

Multi-objective strategy: weighted sum with equal weights. Both costs are in the same monetary unit, so their sum is the real total production cost and needs no arbitrary weights. A lexicographic order would let one criterion dominate (travel savings ignored once actor cost is minimal), and a Pareto front gives many solutions with no natural way to choose one when the goal is a single cheapest schedule.

Results (MiniZinc 2.10.1)

- **Base, Film2**: 87 proved in $\sim$40 s with Chuffed (Gecode: no proof in 2 min, best 130).
- **Extended, Film2 with `travel = 0`**: 87 proved ($\sim$59 s).
- **Extended, `tradeoff_demo`**: 85 proved (0.2 s).
- Film1 and Film2 with travel: best found only, not proved.
- `proj.py` also recomputes each solution's cost in Python and checks it against the solver.

Video

https://youtu.be/nR9Ux0cV56g

How to run

Software: Python 3.12 (tested on 3.12.14), MiniZinc 2.10.1 with Chuffed (bundled), Linux; Gecode is also supported. `minizinc` must be on the `PATH`.

```
python3 proj.py --instance data/tradeoff_demo.json --model extended
python3 proj.py --instance data/film2.json --model base --time-limit 90
```

Options: `-solver` (default `chuffed`), `-time-limit` in seconds (default 60), `-verbose`. The last output line says whether the result is proved optimal. Instances are JSON files in `data/` (format in the README).

Use of web solutions and generative AI

Online models: compared against (not based on) `talent_scheduling_alt.mzn` (MiniZinc benchmarks) and `talent.mzn` (hakank.org); found bug in former ($k < j$ should be $k < l$). Generative AI: Claude Code (Anthropic) used to review models/README, run tests, add CLI options and cost verification to `proj.py`, and organise this document. All results run and checked by us.